```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
print("Libraries imported successfully!")
```

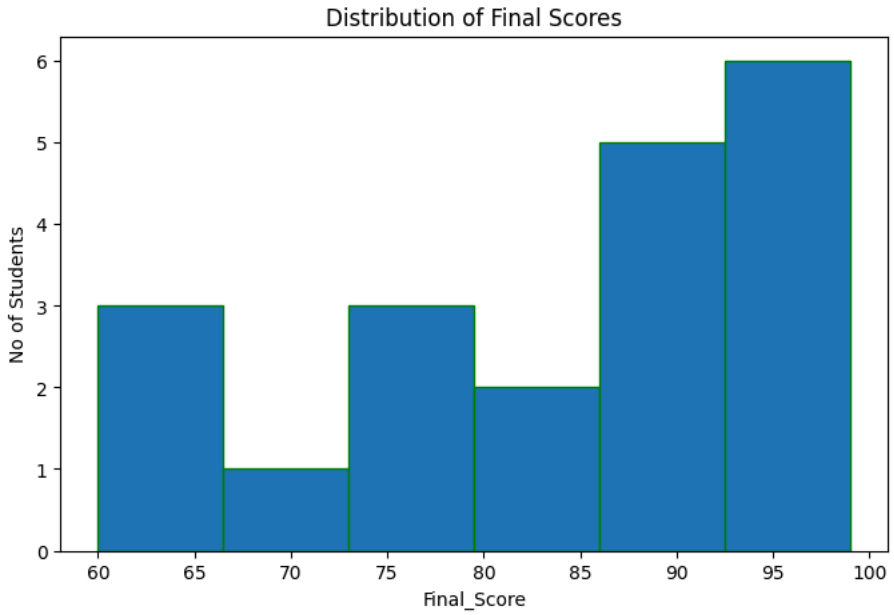
```
Libraries imported successfully!
```

```
data={"Student":["Asha","Bharat","Chitra","Deepak","Esha","Farhan",
                 "Gowri","Harish","Isha","Jatin","Kavya","Lokesh",
                 "Meera","Nikhil","Pooja","Rahul","Sneha","Tarun","Uma","Varun"],
      "Department":["CSE","AIML","CSE","ISE","AIML","CSE","ISE"
      ,"CSE","AIML","ISE","CSE","AIML",
                    "ISE","CSE","AIML","ISE","CSE","AIML","ISE","CSE"],
      "Attendance":[92,78,85,96,88,72,91,65,87,82,94,76,89,97,84,69,93,80,86,90],
      "Study_Hours":[3.5,2,3,5,4,1.5,4.5,1,3.5,2.5,5.5,2,4,6,3,1.5,5,2.5,3.5,4.5],
      "Assignment_Score":[82,68,78,94,86,62,90,55,84,73,96,70,88,98,79,60,93,74,85,91],
      "Internal_Marks":[84,70,80,95,87,65,91,58,85,75,97,72,89,99,81,63,94,76,86,92],
      "Final_Score":[86,72,82,96,88,66,93,60,86,77,98,73,90,99,83,64,95,78,87,94]}
df=pd.DataFrame(data)
display(df)
print("Shape:",df.shape)
print(df.dtypes)
```

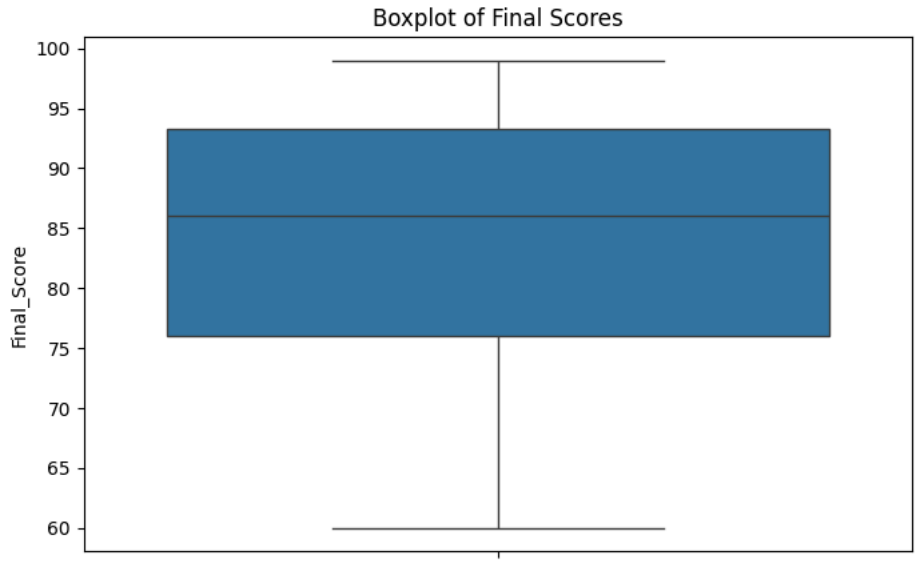
	Student	Department	Attendance	Study_Hours	Assignment_Score	Internal_Marks	Final_Score
0	Asha	CSE	92	3.5	82	84	86
1	Bharat	AIML	78	2.0	68	70	72
2	Chitra	CSE	85	3.0	78	80	82
3	Deepak	ISE	96	5.0	94	95	96
4	Esha	AIML	88	4.0	86	87	88
5	Farhan	CSE	72	1.5	62	65	66
6	Gowri	ISE	91	4.5	90	91	93
7	Harish	CSE	65	1.0	55	58	60
8	Isha	AIML	87	3.5	84	85	86
9	Jatin	ISE	82	2.5	73	75	77
10	Kavya	CSE	94	5.5	96	97	98
11	Lokesh	AIML	76	2.0	70	72	73
12	Meera	ISE	89	4.0	88	89	90
13	Nikhil	CSE	97	6.0	98	99	99
14	Pooja	AIML	84	3.0	79	81	83
15	Rahul	ISE	69	1.5	60	63	64
16	Sneha	CSE	93	5.0	93	94	95
17	Tarun	AIML	80	2.5	74	76	78
18	Uma	ISE	86	3.5	85	86	87
19	Varun	CSE	90	4.5	91	92	94

```
Shape: (20, 7)
Student              object
Department           object
Attendance            int64
Study_Hours         float64
Assignment_Score      int64
Internal_Marks        int64
Final_Score           int64
dtype: object
```

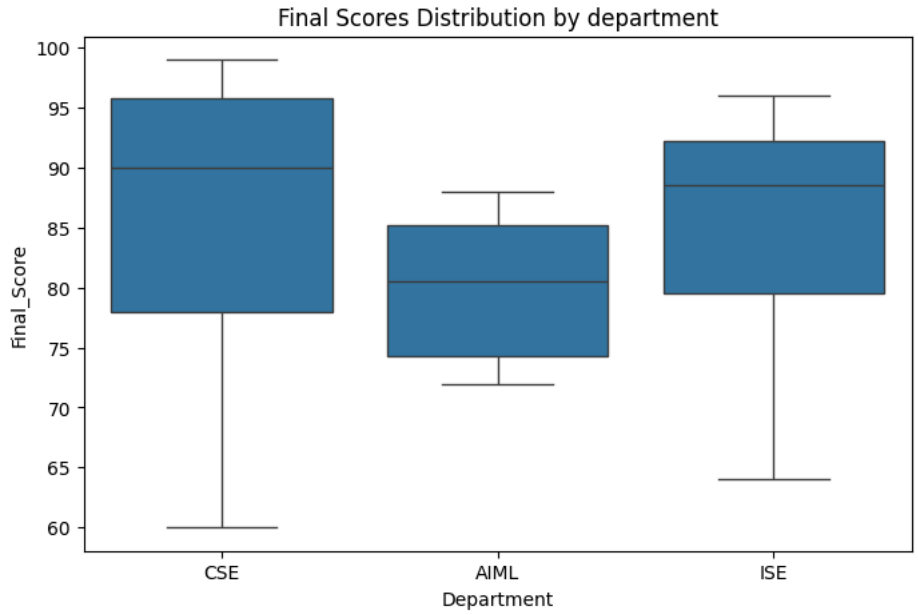
```
plt.figure(figsize=(8,5))
plt.hist(df["Final_Score"],bins=6,edgecolor='green')
plt.title("Distribution of Final Scores")
plt.xlabel("Final_Score");
plt.ylabel("No of Students");
plt.show()
```



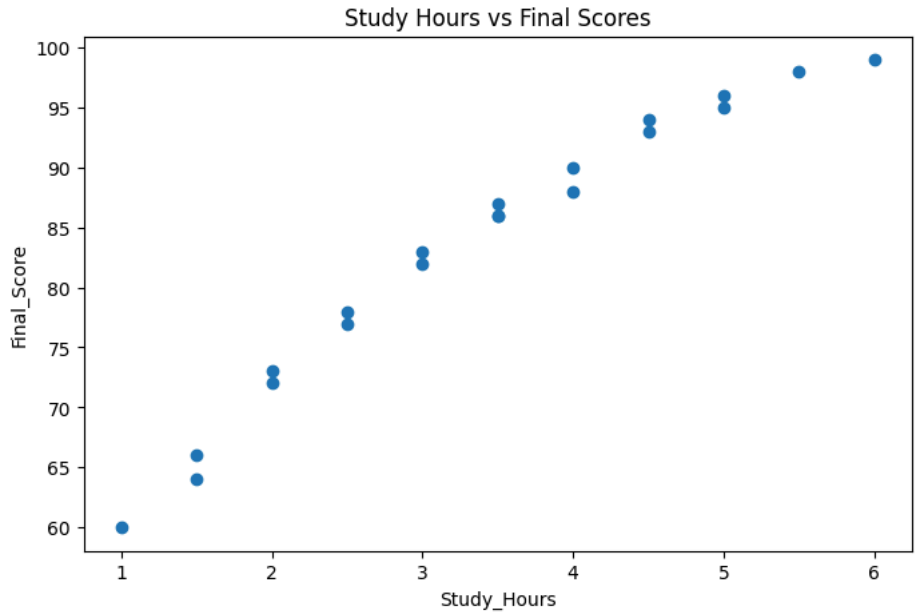

```
plt.figure(figsize=(8,5))
sns.boxplot(y=df["Final_Score"])
plt.title("Boxplot of Final Scores")
plt.show()
```




```
plt.figure(figsize=(8,5))
sns.boxplot(data=df,x="Department",y="Final_Score")
plt.title("Final Scores Distribution by department")
plt.show()
```




```
plt.figure(figsize=(8,5))
plt.scatter(df["Study_Hours"],df["Final_Score"])
plt.title("Study Hours vs Final Scores")
plt.xlabel("Study_Hours");
plt.ylabel("Final_Score");
plt.show()
```




```
plt.figure(figsize=(8,5))
sns.scatterplot(data=df,x="Study_Hours",y="Final_Score",hue="Department",s=100)
plt.title("Study Hours vs Final Scores by department")


plt.show()
```

Show hidden output

```
num_cols=["Attendance","Study_Hours","Assignment_Score","Internal_Marks","Final_Score"]
sns.pairplot(df[num_cols])
plt.show()
```

Show hidden output

```
corr=df[num_cols].corr()
display(corr)
```

Show hidden output

```
plt.figure(figsize=(9,6))
sns.heatmap(corr,annot=True,fmt=".2f",cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

Show hidden output

```
movie_data={"Movie":["Sky Force","Code Red","Midnight Run","Ocean Echo","City Lights","Last Game","Silent Road","Future Wor
            "Budget_Crore":[80,120,60,45,35,70,50,150,40,90,110,130],
            "Duration_Minutes":[145,160,125,135,110,150,130,170,120,155,165,158],
            "Rating":[8.2,7.5,8,7.8,6.9,8.5,7.6,7.2,8.1,7.9,7.4,8.3],
            "Revenue_Crore":[240,280,150,110,75,260,130,310,140,220,290,350]}
movies=pd.DataFrame(movie_data)
movies
```

Show hidden output

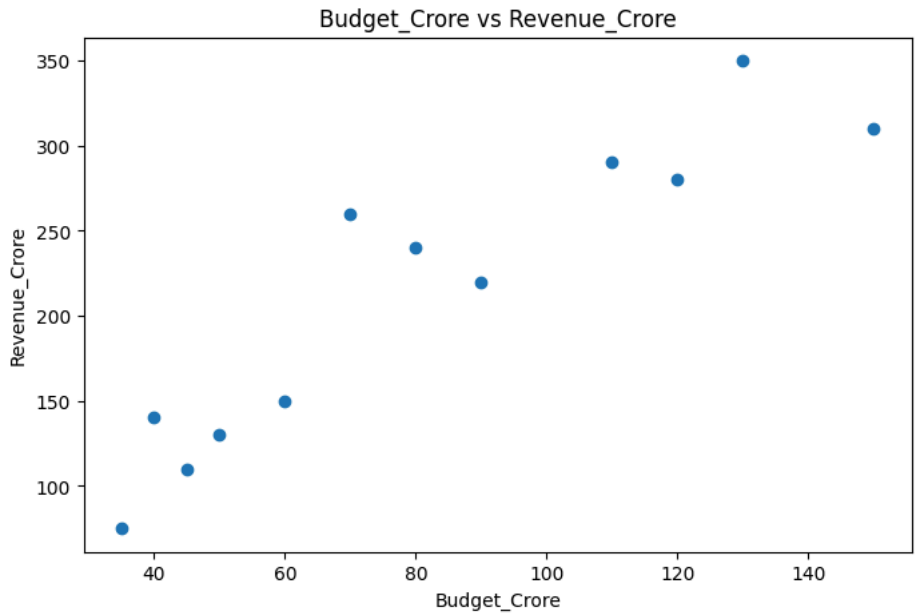
```
plt.figure(figsize=(8,5))
plt.hist(movies["Rating"],bins=6,edgecolor='green')
plt.title("Rating")
plt.xlabel("Rating");
plt.ylabel("Movie");
plt.show()
```

Show hidden output

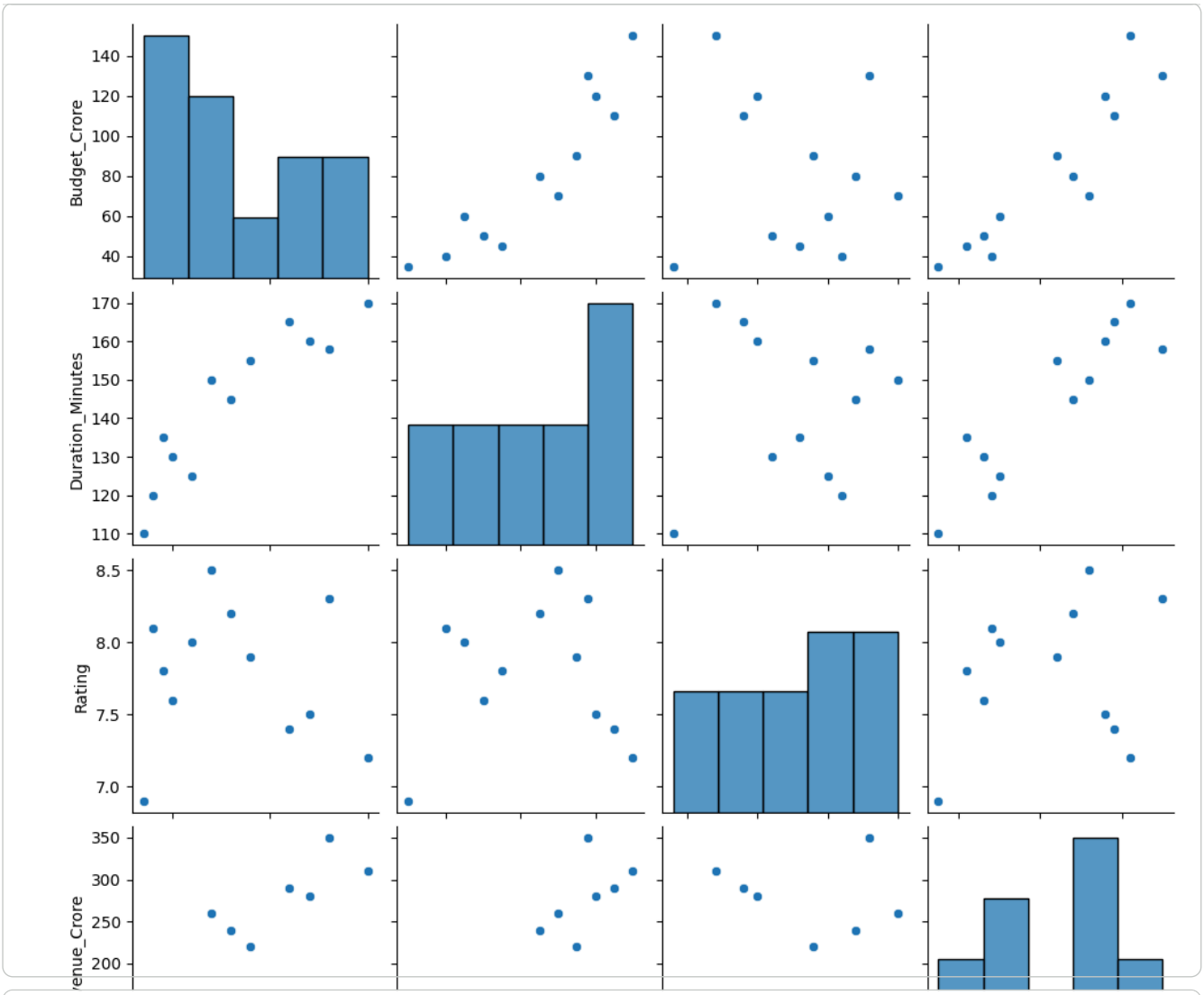
```
plt.figure(figsize=(8,5))
sns.boxplot(y=movies["Budget_Crore"])
plt.title("Boxplot of Budget Crore")
plt.show()
```

Show hidden output

```
plt.figure(figsize=(8,5))
plt.scatter(movies["Budget_Crore"],movies["Revenue_Crore"])
plt.title("Budget_Crore vs Revenue_Crore")
plt.xlabel("Budget_Crore");
plt.ylabel("Revenue_Crore");
plt.show()
```




```
num_cols=["Budget_Crore","Duration_Minutes","Rating","Revenue_Crore"]
sns.pairplot(movies[num_cols])
plt.show()
```




```
corr=movies[num_cols].corr()
display(corr)
```

	Budget_Crore	Duration_Minutes	Rating	Revenue_Crore
Budget_Crore	1.000000	0.914634	-0.080345	0.922029